ENVIRONMENT · CLASS XI

The Ailing Planet: The Green Movement's Role

Understanding Earth's Four Principal Biological Systems and the Consequences of Their Destruction.

Four Biological Systems. One Interconnected Planet. Our Shared Future.

Vaishnavi Kashyap & Rani Singh, Class XI — Mathematics





FISHERIES

Fisheries: the living blue economy

Marine and inland capture fisheries provide food and livelihoods; aquaculture farms aquatic organisms. Healthy food webs and coastal nurseries sustain wild stocks.

Overfishing, bycatch, destructive gear and pollution weaken resilience. Science-based quotas, protected breeding grounds, monitoring and responsible aquaculture help.

62.3%

of assessed marine stocks were biologically sustainable in 2021, down from 90% in 1974.

01

Habitats

02

Fish populations

03

Livelihoods

FORESTS

Forests: the planet's living archive

Forests shelter biodiversity, store carbon and support livelihoods through timber, fuel, food and medicines. They are part of the climate answer, not a solution alone.

4.14 billion ha

Forests cover 32% of land.

4.12 million ha

Annual net forest loss fell from 10.7 million ha (1990–2000) to 4.12 million ha (2015–2025).

Vegetation + soil

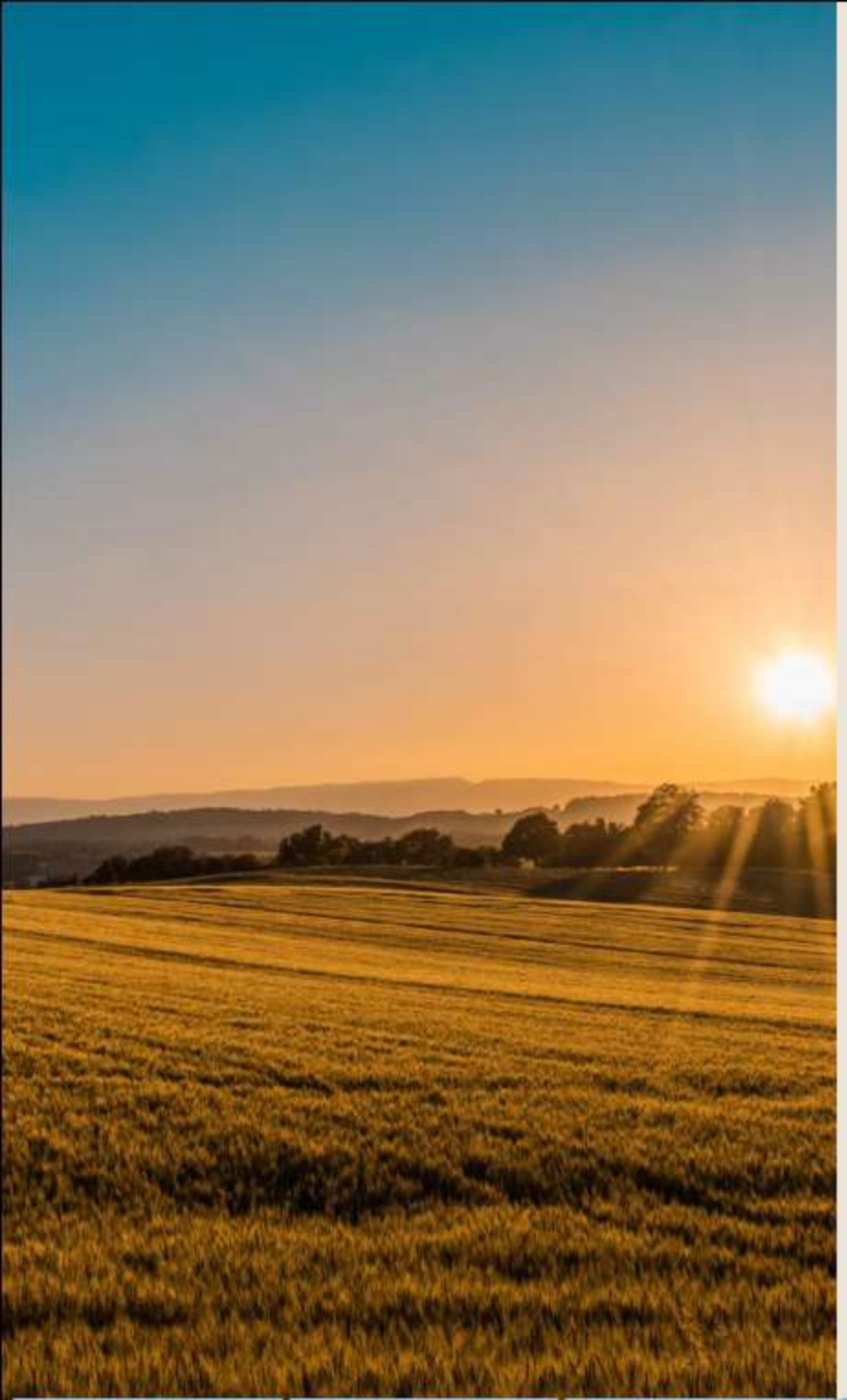
Roots protect soil; soil sustains growth.

Water + biodiversity

Watersheds and habitats sustain life.

Deforestation, unsustainable logging, farm expansion, mining, infrastructure and severe fires demand conservation, responsible management and restoration.

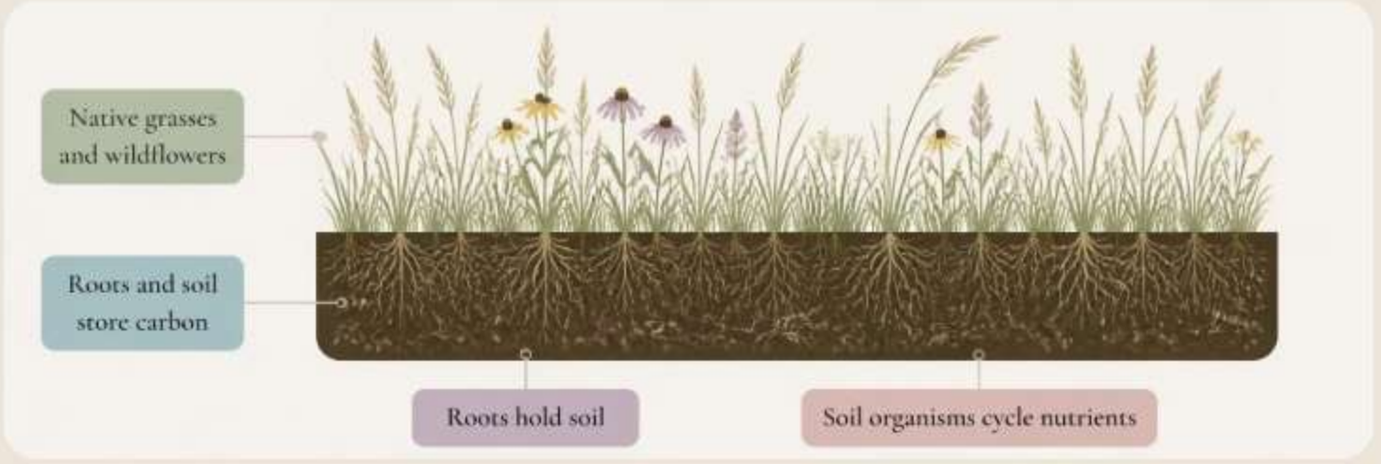




GRASSLANDS

Grasslands: the ecosystems we often overlook

Native grasses, wildflowers, insects, birds and grazing animals form rich communities. Much of their carbon lies below ground in roots and soil.

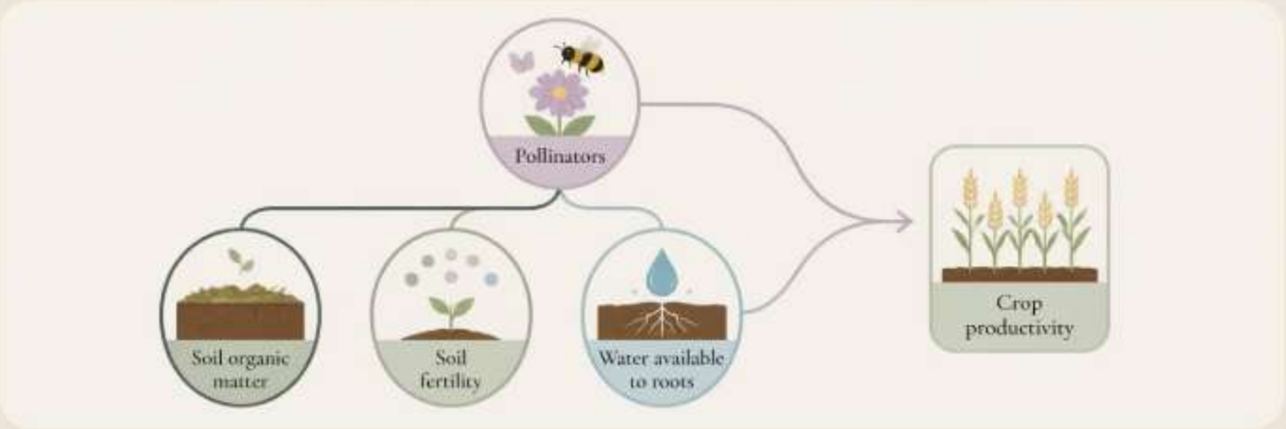


Sustainable grazing supports pastoral livelihoods. Conversion, overgrazing, invasive species and altered fire regimes cause degradation.

Natural grasslands are not degraded land awaiting tree planting.

Croplands: the foundation of food security

Croplands grow cereals, pulses, fruits, vegetables and oilseeds for food, fibre and feed; they support food security, employment, trade and industrial raw materials.



Threats include erosion, nutrient depletion, salinity, poor irrigation, compaction and mismanaged agrochemicals. Context-suited responses: crop rotation, soil conservation, integrated pest management, efficient water use and agroforestry.

Agricultural land includes cropland and permanent pasture.





Cash crops: economic value, ecological responsibility

Cash crops are grown primarily for sale: cotton, tea, coffee, sugarcane and jute. They belong within croplands, not a fifth biological system.

Economic contribution

Tea supports farmer income, jobs and trade. Other cash crops supply textiles, beverages and sugar.

Conditions for sustainability

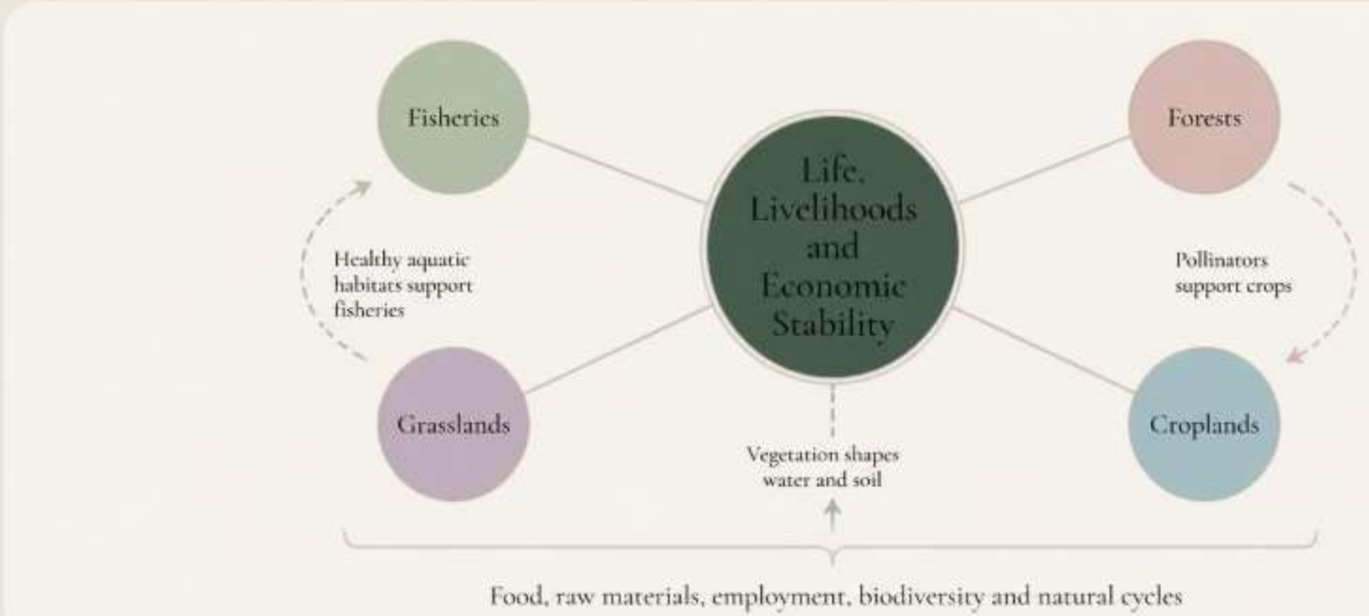
Locally suited planning, soil protection, integrated pest management and traceable supply chains limit ecological harm.

Monoculture, conversion, water use and inputs require care; not every cash crop is unsustainable.



ONE INTERCONNECTED PLANET

Why all four biological systems matter



Soil, water, nutrients and carbon connect these systems. Together, they support well-being and resilience.

When natural systems lose productivity, the damage extends beyond nature to food security, livelihoods and the global economy.

PRESSURE AND LIMITS

Present threats: why Earth's systems are under pressure

System	Main pressure
Fisheries	Overfishing
Forests	Deforestation and habitat conversion
Grasslands	Degradation and overgrazing
Croplands	Soil degradation, water over-extraction and inputs

Pollution, climate change and biodiversity loss compound these pressures. Using renewables faster than they replenish reduces long-term productivity.

Consumption, production methods, technology, institutions and inequality matter alongside population.

37.7%

Assessed marine stocks unsustainable, 2021

4.12 million ha/year

Net forest loss, 2015–2025



FROM DOMINATION TO PARTNERSHIP

When nature loses its balance, humanity feels the consequences

Falling yields, biodiversity loss, erosion, disrupted water and desertification in vulnerable regions reinforce economic insecurity and weaken resilience.

Protect and restore ecosystems; fish and log responsibly; care for soil and water; use renewables within limits; reduce waste; strengthen governance and participation.

Sustainable development meets present needs without compromising future generations. Palkhivala calls for partnership with nature.

“The future of the planet is inseparable from the choices we make today.”

Original closing line

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Sources: NCERT, Nani Palkhivala's chapter · FAO SOFIA 2024 · FAO FRA 2025
Data years: 2021; 2025

